Introducing Extreme Value Theory

Edinburgh Summer School on Generative AI for Extremes

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University of Edinburgh

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JCMB, Edinburgh

THE UNIVERSITY of EDINBURGH

una europa

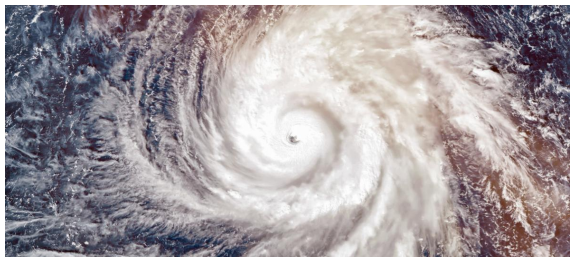
Introducing Myself

Briefing

Credentials

- Professor at the School of Mathematics, UoE
- Chair of Statistical Data Science
- Co-Director, Edinburgh Centre for Financial Innovations
- Elected Fellow, Generative AI Laboratory

Leading research field: Extreme Value Theory.



Other interests: Bayes; Interfaces between Statistics & AI.

Recent Funding

Research bodies

LEVERHULME TRUST

RSE The Royal Society of Edinburgh KNOWLEDGE MADE USEFUL

ℙrob_AI

Industry

aberdeen Investments

Unilever

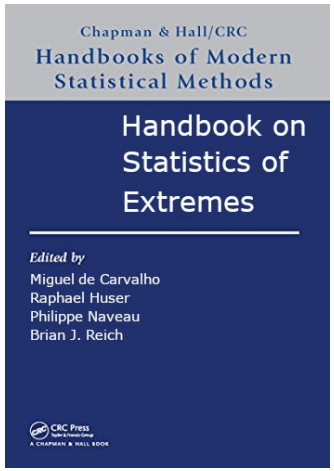
Introducing Myself

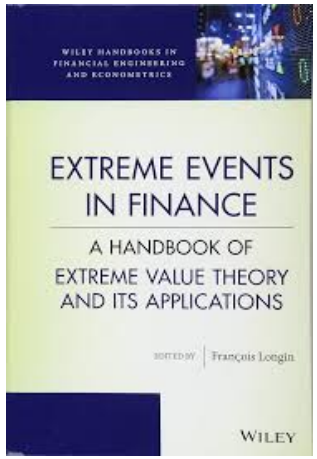
GAME 2025

Extremes, Special Issue (2026)

EVA 2021

Handbooks





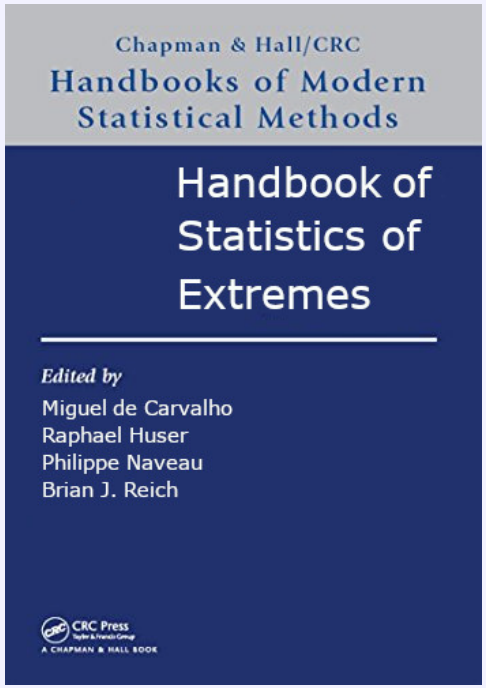
Agenda

- Introduction
- Block Maxima and the Generalized Extreme Value Distribution
- Threshold Exceedances and the Generalized Pareto Model
- Changing Environments and Temporal Extremes
- A Primer on Multivariate Extremes
- Conclusion
- EVT and AI: Parallel Beginnings

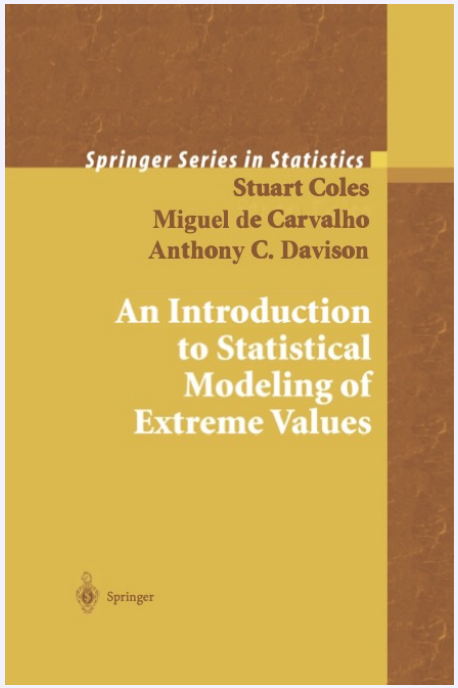
Sources for These Notes

Background Reading

Parts of these notes are based on the following two books.



Handbook
(in print; expected 2026)



Coles et al.
(expected 2027)

Statistics of Extremes

Context and Applied Motivation

Statistics of extremes:

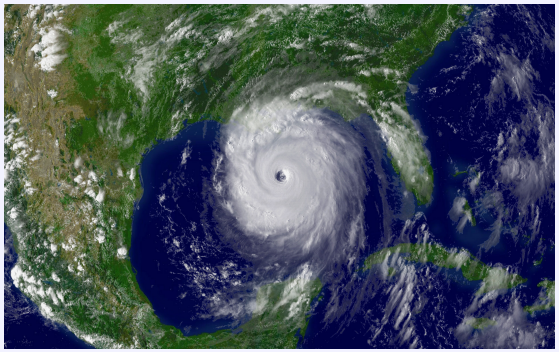
- Modelling extreme events and quantifying their likelihood of occurrence.

Applied motivation:

- "Are the losses induced by Hurricane Katrina a one-in-100-year event?"
- "How likely is it for a portfolio of crypto-assets to suffer a joint crash?"
- "How often should we expect storm surges inducing extreme sea levels?"

In Cauda Venenum

The Poison in the Tail



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TUESDAY, SEPTEMBER 16, 2008 - VOL. CCLII NO. 65

AIG, Lehman Shock Hits World Markets

Focus Moves to Fate of Giant Insurer After U.S. Allows Investment Bank to Fail; Barclays in Talks to Buy Core Lehman Unit

AIG Faces Cash Crisis As Stock Dives 61%

Floods in Australia, wildfires in Portugal, earthquake / tsunami in Japan, hurricane Katrina, mudslides in Rio de Janeiro, stock market crash.

Statistical Questions

Severity, Frequency and Simultaneous Extremes

Example (A Coastal Storm)

How high might the next extreme sea level be? How often is a critical level exceeded? How likely are extreme sea level and heavy rainfall to occur together?

For variables X and Y, with marginal distribution functions F_X and F_Y, typical targets are

$$P(X > x), \qquad F_X^{-1}(1-p), \qquad P(X > x, Y > y),$$

for high levels x, y and small p.

Key Point

Extreme value theory (EVT) supplies structured approximations for the part of a distribution where observations are scarce.

A Brief History of Extreme Value Theory

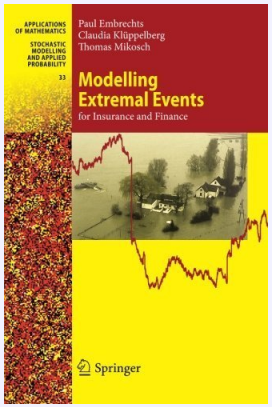
Timeline

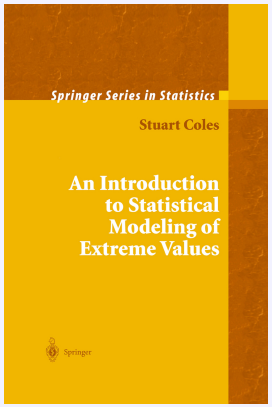
1920s	Asymptotics for maxima—foundations
1940s	Asymptotics for maxima—refinements
1950s	Statistical modeling, I
1970s	Asymptotics for exceedances
1980s	Asymptotics for stationary processes
1990s	Nonstationary extremes and joint extremes
2000s	Statistical modeling, II
2010s	Space–time extremes and nonstationary joint extremes
2020s	High-dimensional extremes and interfaces with machine learning

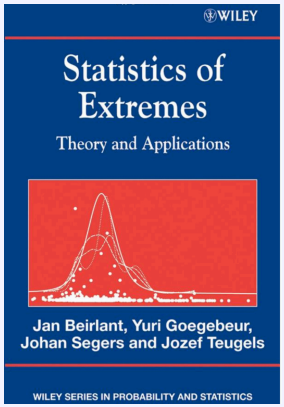
Books and Review Papers

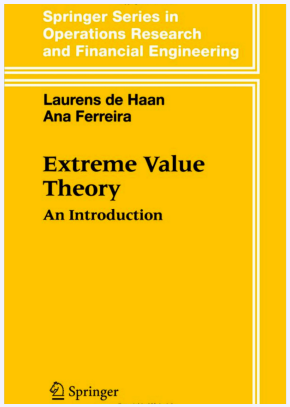
Selected Resources

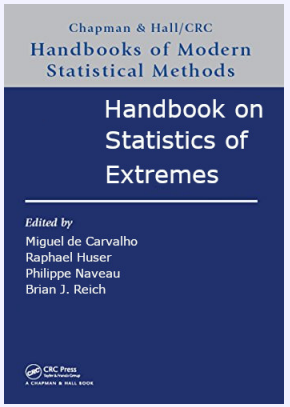
Books (examples):











Review papers (examples):

- Davison, A. C., and Huser, R. (2015), "Statistics of Extremes," *Annual Review of Statistics and Its Application*, 2, 203–235.
- Huser, R., and Wadsworth, J. L. (2022), "Advances in Statistical Modeling of Spatial Extremes," *WIREs Computational Statistics*, 14(1), e1537.

Software and Journal

Resources for Practice and Research

Software ecosystems:

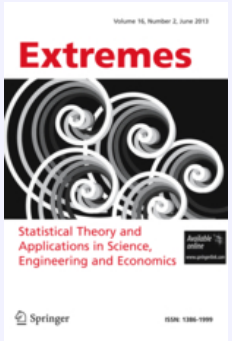
- **R:** CRAN Task View on Extreme Value Analysis: `https://cran.r-project.org/web/views/ExtremeValue.html`
- **Julia:** *Extremes.jl*; **Python:** *pyextremes*.

Software review:

- Belzile, L. R., Dutang, C., Northrop, P. J., and Opitz, T. (2023), "A Modeler's Guide to Extreme Value Software," *Extremes*, 26, 595–638.

Journal:

- *Extremes* (Springer; Journal Impact Factor 1.9, 2025).



Meetings

The Extreme Value Analysis Conference

EVA 2021, Edinburgh

Online, 28 June–2 July 2021

EVA 2023, Milan

Bocconi University, 26–30 June 2023

EVA 2025, North Carolina

UNC Chapel Hill, 23–27 June 2025

EVA 2027, Montréal

Next: HEC Montréal, 5–9 July 2027

Black Swans

The Empirical Distribution Cannot Extrapolate

- Consider a random sample

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} F.$$

The empirical distribution function is

$$\widehat{F}_n(x) = \frac{1}{n} \sum_{i=1}^{n} I(X_i \leq x).$$

- The empirical distribution function cannot extrapolate beyond the observed data. For every $\varepsilon > 0$,

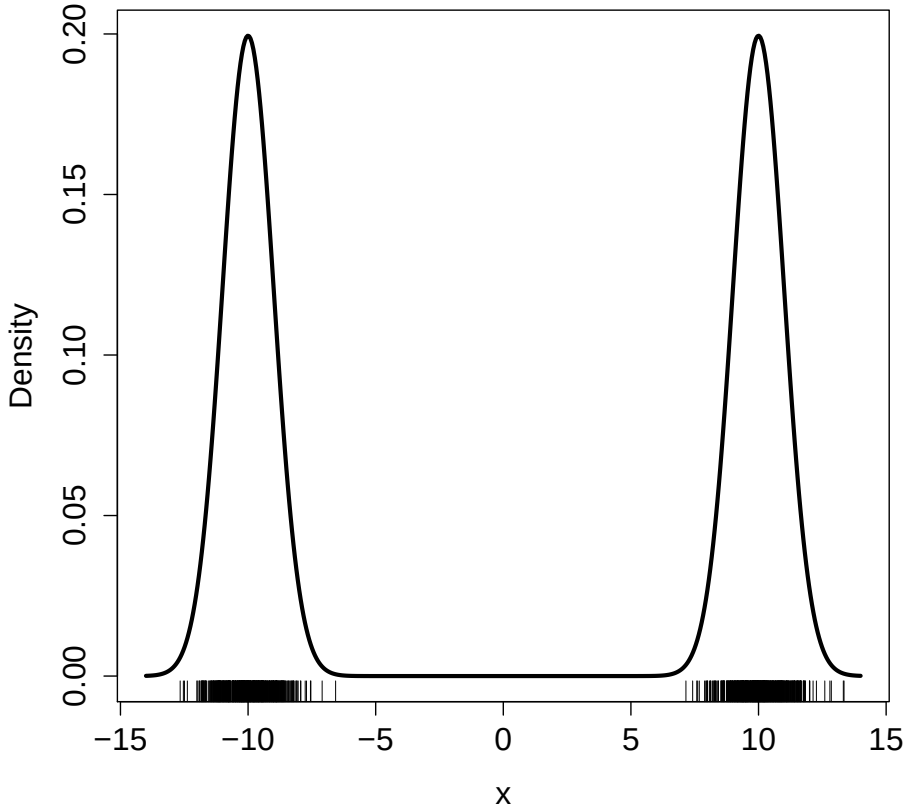
$$1 - \widehat{F}_n(M_n + \varepsilon) = 0,$$

where $M_n = \max\{X_1, \ldots, X_n\}$.

Extreme $\neq$ Rare

Rarity Is Not Extremeness

- Statistics of Extremes is about small probability events, linked to the tail of a distribution.
- But beware, rare is not the same as extreme.



rare and extreme rare rare and extreme

Why Extreme Value Theory?

Fitting the Bulk and Extrapolating the Tail Are Different Problems

- **Empirical and flexible models** can describe common observations well.
- **Extreme-value questions** concern levels with little or no direct information in the data.
- **EVT** uses limiting arguments to motivate tail families, return levels and extremal dependence models.

Example (Assessing storm risk)

Matching the bulk of observed storm intensities does not establish reliable estimates of very small exceedance probabilities or simultaneous extremes across locations.

Maxima

Sample Maxima and the Right Endpoint

Maxima:

- Maxima are natural for modeling extreme events.
- Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} F$ be a random sample. The sample maximum is

$$M_n = \max\{X_1, \ldots, X_n\}.$$

- Following Coles, write the right endpoint as

$$x_+ = \sup\{x : F(x) < 1\};$$

it is the population counterpart of the sample maximum.

The Law of the Maximum

Without Normalization, the Limit Is Degenerate

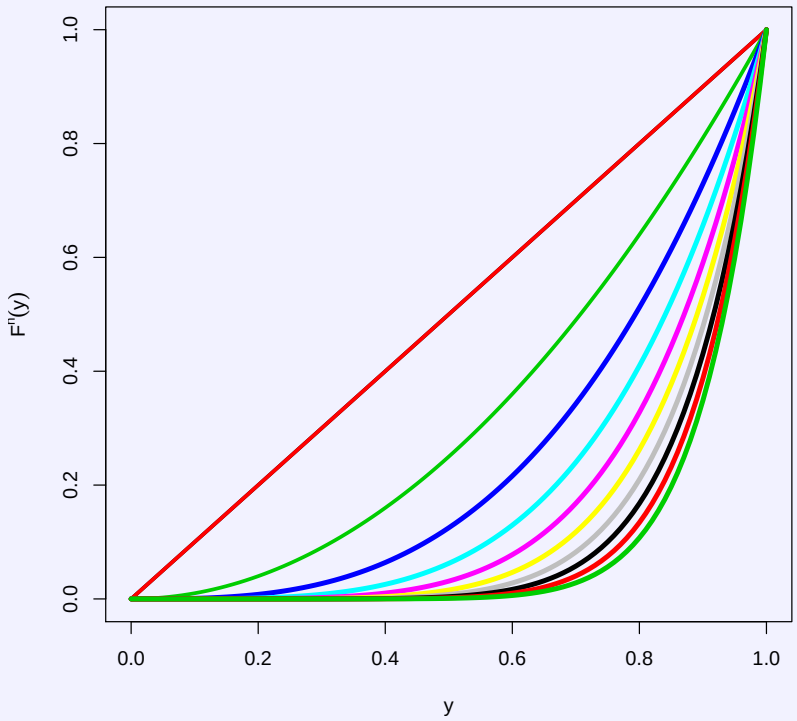
For independent observations:

$$P(M_n \leq z) = P(X_1 \leq z, \ldots, X_n \leq z)$$
$$= \prod_{i=1}^{n} P(X_i \leq z)$$
$$= F^n(z).$$

Thus, as $n \to \infty$:

- $F^n(z) \to 0$ for every fixed $z < x_+$.
- If $x_+ < \infty$, then $F^n(z) = 1$ for $z \geq x_+$.

Example: $X \sim \text{Unif}(0, 1)$



A Weak Law for Maxima

The Sample Maximum Approaches the Right Endpoint

Theorem (Weak Law for Maxima)

For the right endpoint $x_+ \in \mathbb{R} \cup \{\infty\}$,

$$M_n \xrightarrow{P} x_+.$$

Proof.

If $x_+ < \infty$, then $M_n \leq x_+$ almost surely and $F(x_+ - \varepsilon) < 1$ for every $\varepsilon > 0$. Hence

$$P(|M_n - x_+| \geq \varepsilon) = P(M_n \leq x_+ - \varepsilon) = F^n(x_+ - \varepsilon) \longrightarrow 0.$$

If $x_+ = \infty$, then $F(x) < 1$ for every finite x, and thus

$$P(M_n > x) = 1 - F^n(x) \longrightarrow 1.$$

□

The Normal-Distribution Analogy

Limit Laws for Averages and for Maxima

Averages

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

For iid data with $\mathrm{E}\,|X_i| < \infty$ and mean m, the weak law gives

$$\overline{X}_n \xrightarrow{P} m.$$

If $\mathrm{var}(X) < \infty$, the central limit theorem gives

$$\frac{\sqrt{n}(\overline{X}_n - m)}{\sqrt{\mathrm{var}(X)}} \xrightarrow{d} \mathrm{N}(0,1).$$

Maxima

$$M_n = \max_{1 \le i \le n} X_i$$

For finite right endpoint x_+, the weak law for maxima gives

$$M_n \xrightarrow{P} x_+.$$

For suitable $a_n > 0, b_n$, an extreme-value limit has

$$M_n^* = \frac{M_n - b_n}{a_n}, \qquad M_n^* \xrightarrow{d} G_\xi.$$

The GEV is a counterpart to the normal distribution for extremes. The assumptions and normalization are different.

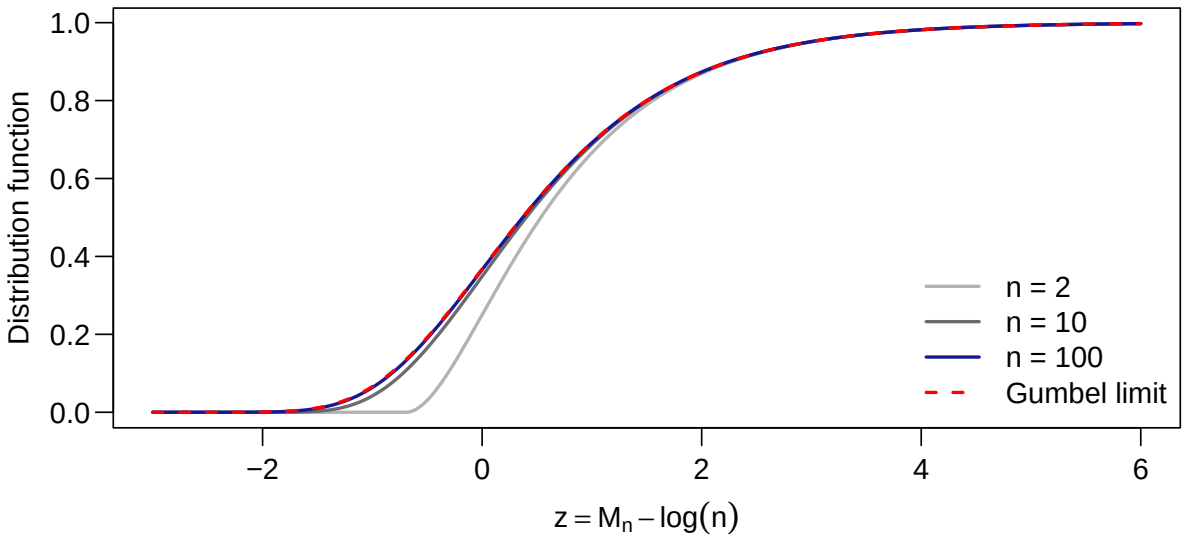
A Limit Law We Can Derive

Exponential Observations Lead to a Gumbel Maximum

Example ($X_i \overset{\text{iid}}{\sim} \text{Exp}(1)$)

For fixed z and sufficiently large n,

$$P(M_n - \log n \leq z) = \left(1 - \frac{e^{-z}}{n}\right)^n \longrightarrow \exp(-e^{-z}).$$



The Extremal Types Theorem

The Possible Nondegenerate Limits Form One Family

Candidate tail models must be capable of extrapolating beyond the observed data.

Theorem (Fisher–Tippett–Gnedenko)

If $X_i \overset{\text{iid}}{\sim} F$ *and, for some* $a_n > 0, b_n$,

$$P\{M_n^* \leq z\} \longrightarrow G_\xi(z), \qquad M_n^* = \frac{M_n - b_n}{a_n}$$

at continuity points of a nondegenerate G_ξ*, then* G_ξ *is a GEV distribution, up to location and scale.*

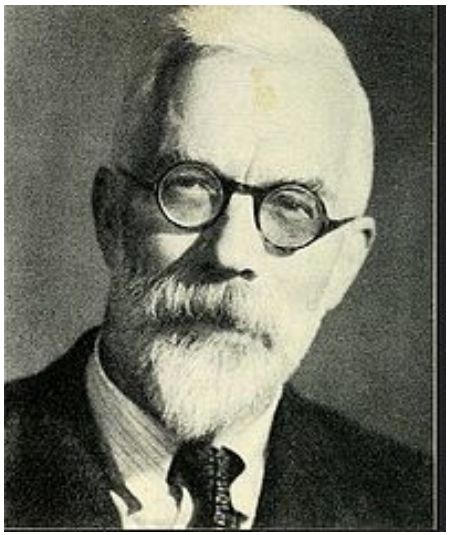
In standardized form,

$$G_\xi(z) = \exp\{-(1+\xi z)^{-1/\xi}\}, \qquad 1+\xi z > 0,$$

with $G_0(z) = \exp(-e^{-z})$. The theorem characterizes possible limits; it does not assert convergence for every F, or specify an adequate finite block length.

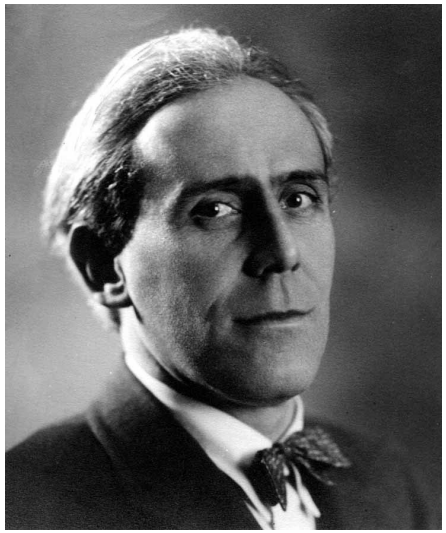
The Trinity of Limiting Distributions

Gumbel, Fréchet and Reverse Weibull


R. A. Fisher

B. Gnedenko

M. Fréchet


E. Gumbel

Notes:

- μ and σ are location and scale parameters; ξ is a shape parameter that determines the limiting upper-tail regime and support:
  1. $\xi = 0$: Gumbel domain;
  2. $\xi > 0$: heavy-tailed Fréchet domain;
  3. $\xi < 0$: reverse-Weibull domain with a finite upper endpoint.

The Generalized Extreme Value Distribution

A Statistical Model for Block Maxima

The three limiting distributions can be combined into the generalized extreme value family:

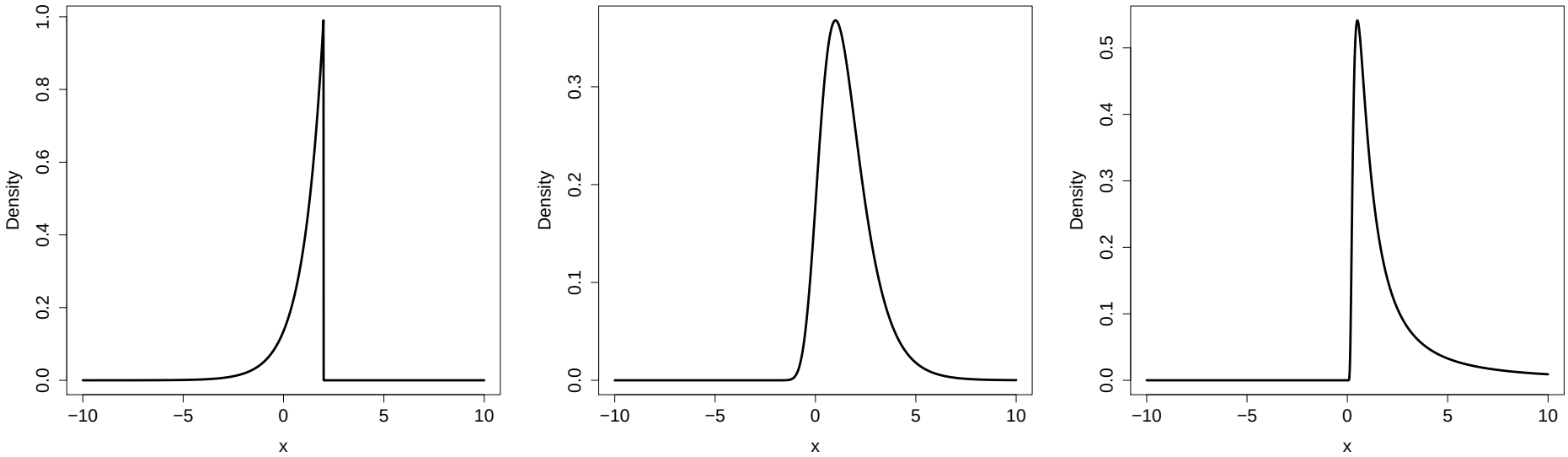
$$G(z) = \exp\left[-\left\{1+\xi\frac{z-\mu}{\sigma}\right\}^{-1/\xi}\right], \quad 1+\xi\frac{z-\mu}{\sigma} > 0.$$

- $\mu \in \mathbb{R}$: location; $\sigma > 0$: scale; $\xi \in \mathbb{R}$: shape.
- At $\xi = 0$, $G(z) = \exp[-\exp\{-(z-\mu)/\sigma\}]$.
- If $\xi > 0$, $G(z) = 0$ below the lower endpoint $\mu - \sigma/\xi$.
- If $\xi < 0$, $G(z) = 1$ above the upper endpoint $\mu - \sigma/\xi$.

The unknown normalization is absorbed into the fitted location and scale.

Generalized Extreme Value Densities

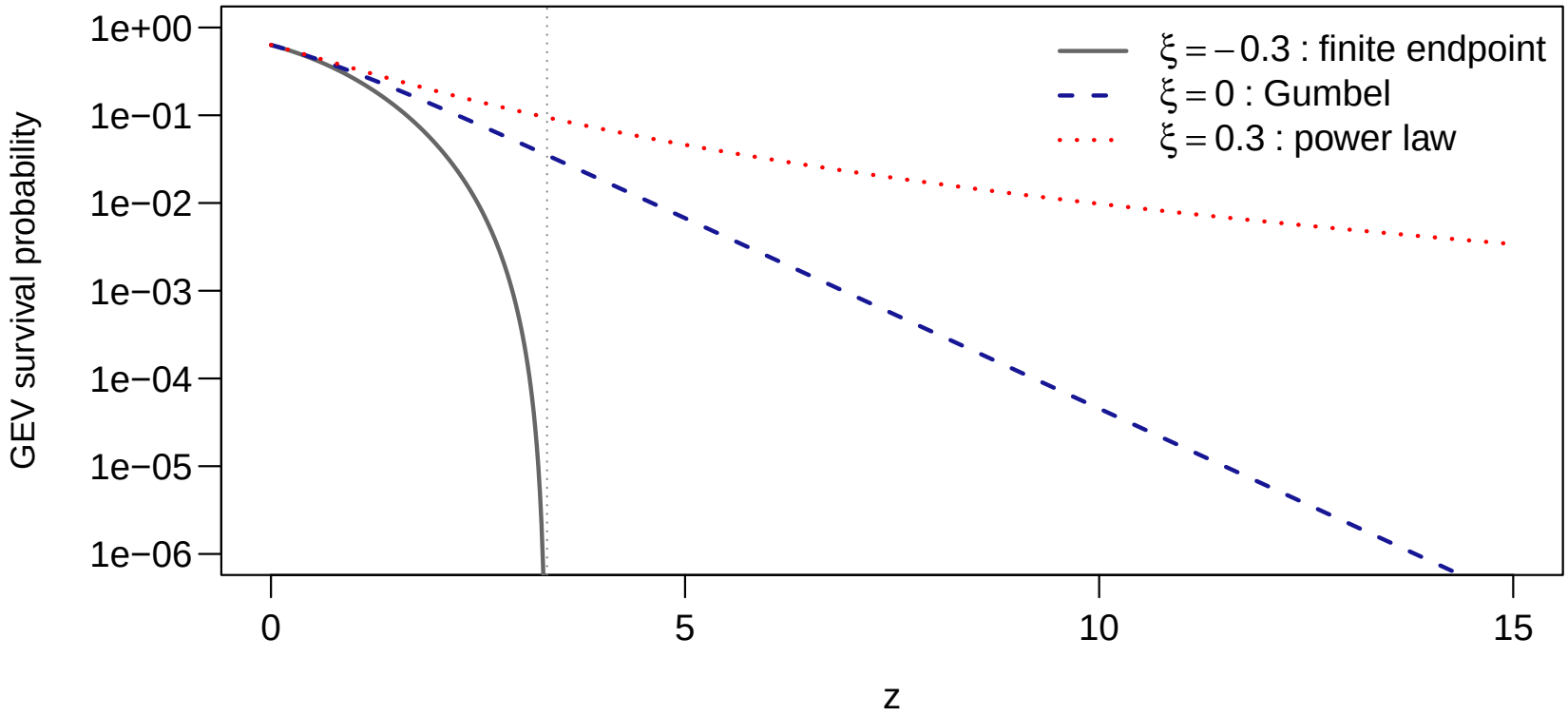
Three Shapes with Common Location and Scale



Densities of GEV$(1, 1, -1)$, GEV$(1, 1, 0)$, and GEV$(1, 1, 1)$; reverse Weibull, Gumbel, and Fréchet, respectively.

Three Tail Regimes

The Shape Parameter Controls Extrapolation



- $\xi > 0$: power-law upper tail (Fréchet).
- $\xi = 0$: Gumbel; includes maxima of normal and exponential data.
- $\xi < 0$: finite upper endpoint (reverse Weibull).

Tail Behaviour and Moments

The Extreme Value Index Is Not a Variance Parameter

For a heavy-tailed parent in the Fréchet domain,

$$\frac{1-F(tx)}{1-F(t)} \longrightarrow x^{-1/\xi}, \qquad x>0, \quad \xi>0.$$

This is **regular variation**; $\alpha = 1/\xi$ is a common tail-index convention.

Example (A Pareto tail)

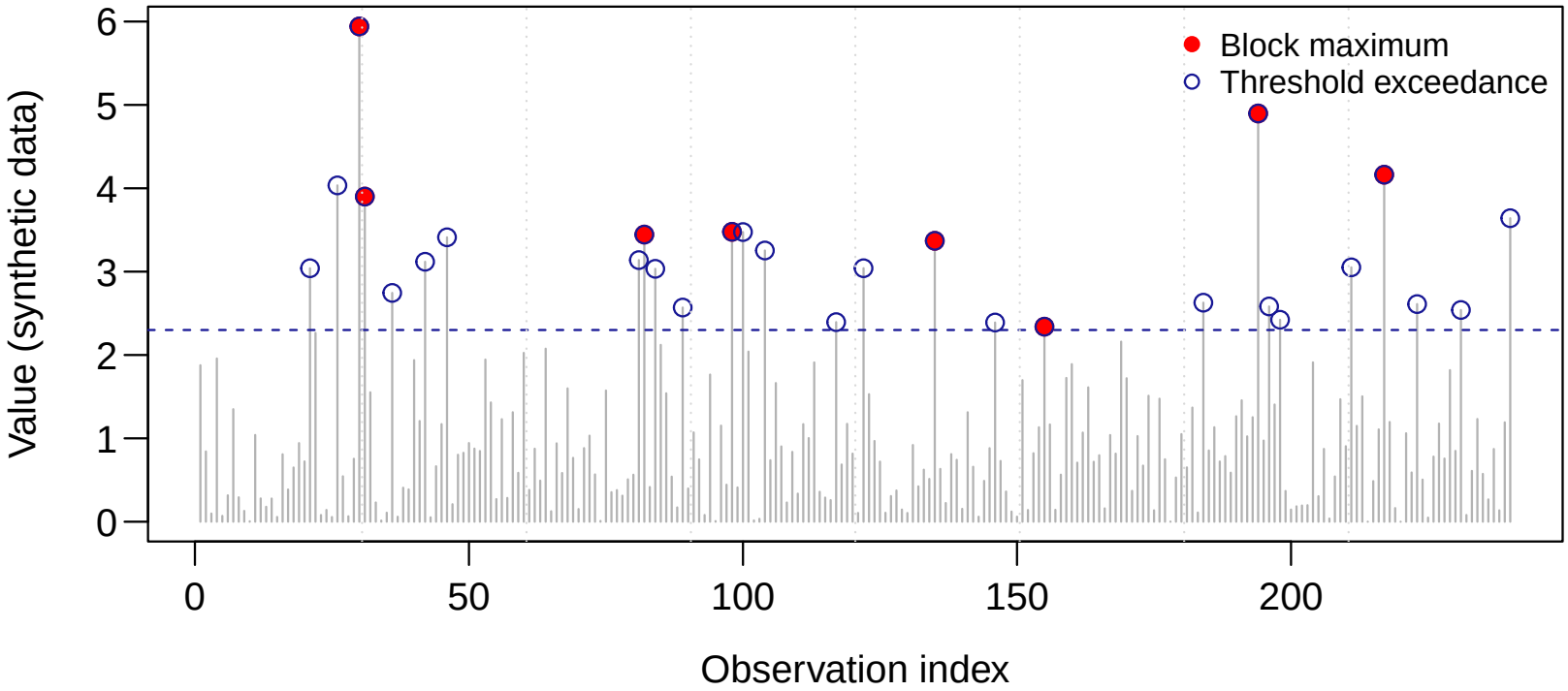
If $P(X>x) = x^{-\alpha}$ for $x \geq 1$, then $\xi = 1/\alpha$ and $\mathrm{E}(X^r) < \infty$ exactly when $r < \alpha$, for $r > 0$.

For a GEV variable, the mean exists if $\xi < 1$ and the variance if $\xi < 1/2$.

High quantiles remain well-defined even when moments do not exist.

Block Maxima in Practice

The Sampling Unit Changes When We Extract Extremes



With m blocks of length n, retain

$$M_{n,j} = \max\{X_{(j-1)n+1}, \ldots, X_{jn}\}, \quad j = 1, \ldots, m.$$

Larger blocks may improve the approximation but leave fewer maxima for inference.

Return Levels

A Quantile Indexed by Its Exceedance Probability

If G is the distribution function of annual maxima, the return level z_p satisfies

$$G(z_p) = 1 - p, \qquad 0 < p < 1,$$

and has return period $1/p$ years. For the GEV distribution,

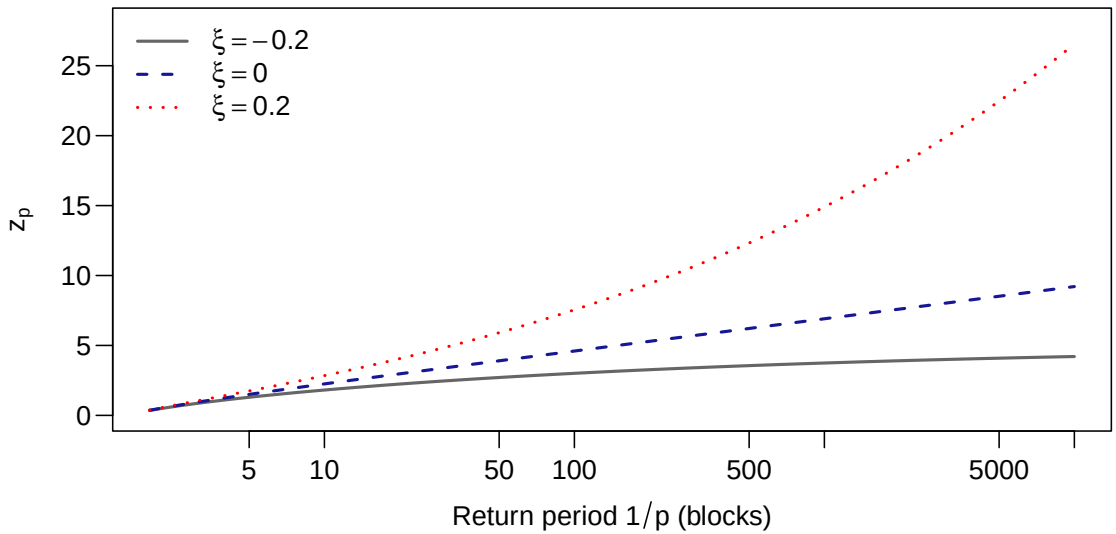
$$z_p = \begin{cases} \mu - \dfrac{\sigma}{\xi}\left[1 - \{-\log(1-p)\}^{-\xi}\right], & \xi \neq 0, \\ \mu - \sigma \log\{-\log(1-p)\}, & \xi = 0. \end{cases}$$

Example (A 100-Year Level)

Here $p = 0.01$, so the return level is $z_{0.01}$. With iid annual maxima, the chance of at least one exceedance in 100 years is $1 - 0.99^{100} \approx 0.634$.

Small Changes in Shape, Large Changes in Risk

Equal Location and Scale Do Not Imply Similar Return Levels



All curves use $\mu = 0, \sigma = 1$. They differ only in ξ.

Likelihood for GEV Models

Learning from Block Maxima

For independent block maxima $z_1, \ldots, z_m$, let

$$t_j = 1 + \xi \frac{z_j - \mu}{\sigma} > 0.$$

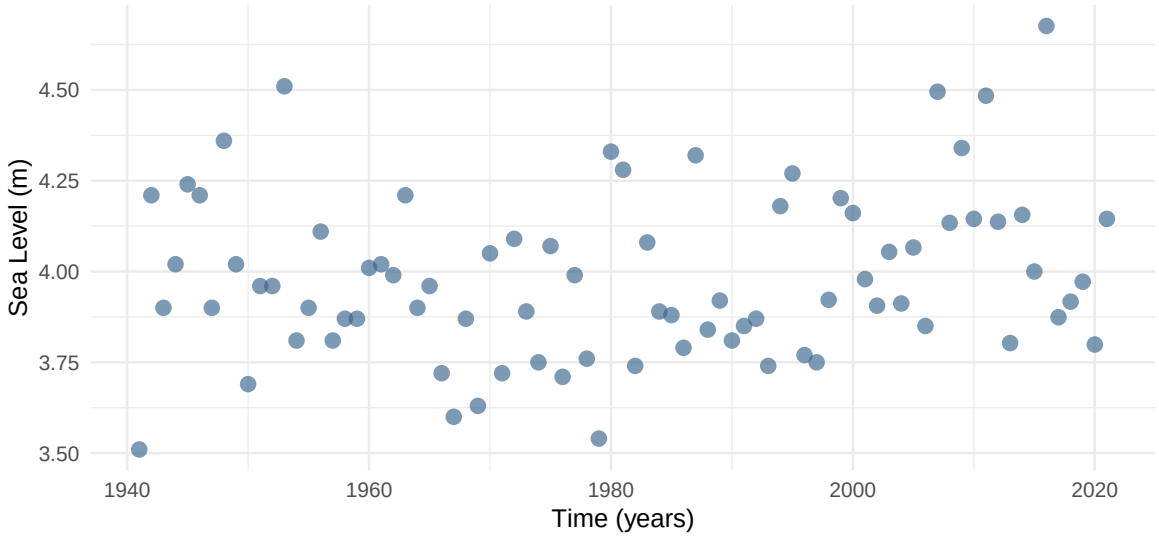
For $\xi \neq 0$, the log-likelihood is

$$\ell(\mu, \sigma, \xi) = -m \log \sigma - \left(1 + \frac{1}{\xi}\right) \sum_{j=1}^{m} \log t_j - \sum_{j=1}^{m} t_j^{-1/\xi}.$$

Numerical maximization gives $(\widehat{\mu}, \widehat{\sigma}, \widehat{\xi})$; the support constraint must hold for every observed maximum.

Port Pirie Annual Maxima

A Real GEV Analysis Using `ismev2`



Data: 81 annual maximum sea levels, 1941–2021.

Stationary GEV fit:

$$\widehat{\mu} = 3.890, \quad \widehat{\sigma} = 0.202,$$

$$\widehat{\xi} = -0.123.$$

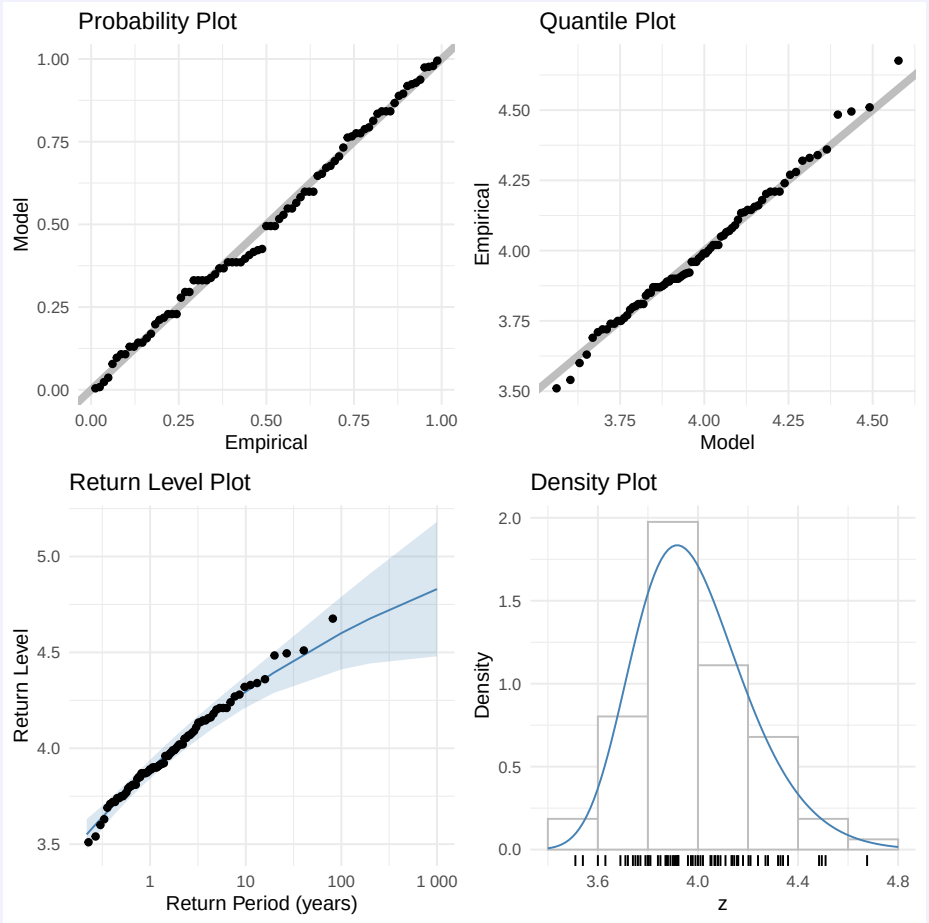
100-year return level ($p = 0.01$):

$$\widehat{z}_{0.01} = 4.600 \text{ m}.$$

```
fit <- gev.fit(portpirie$SeaLevel)
```

Port Pirie Annual Maxima

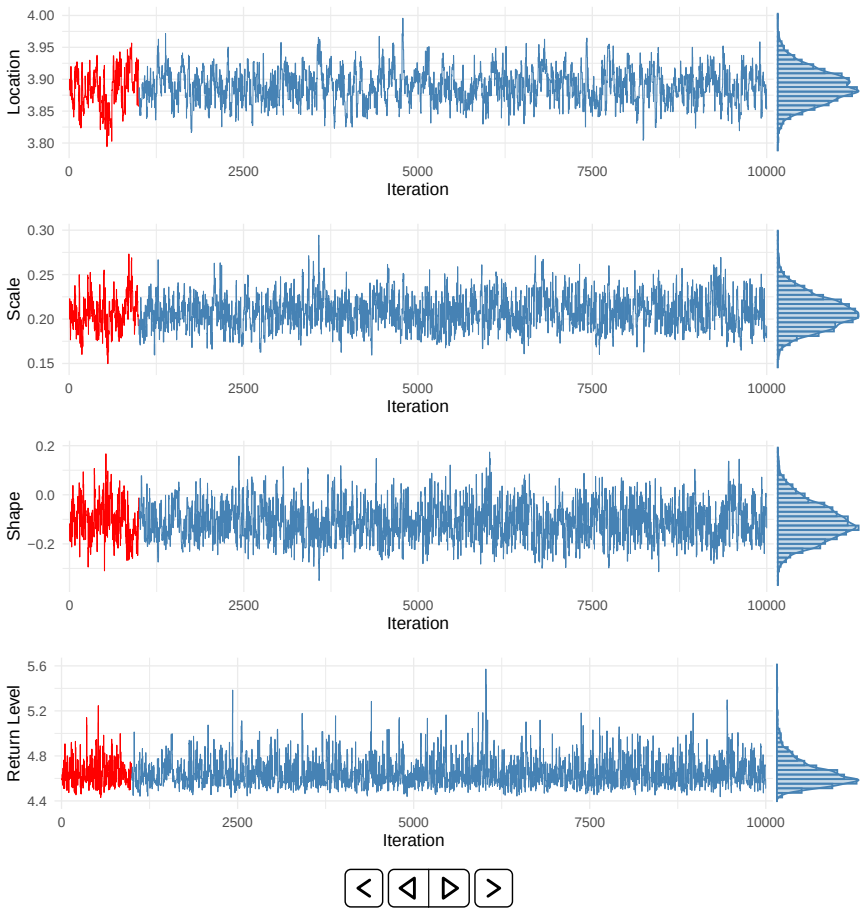
Diagnostics from `ismev2`



Probability, quantile, return-level and density views probe different aspects of fit.

Port Pirie Annual Maxima

A Bayesian GEV Analysis as the Markov Chain Runs



The same **GEV likelihood** from `ismev2` can be combined with a prior:

$$\boldsymbol{\eta} = (\mu, \log \sigma, \xi),$$
$$\pi(\boldsymbol{\eta} \mid \mathbf{z}) \propto L(\boldsymbol{\eta}; \mathbf{z}) \pi(\boldsymbol{\eta}).$$

Red: first 1,000 iterations.

Blue: 9,000 retained iterations.

Transforming every retained draw gives the 100-year return level $z_{0.01}$: posterior mean 4.646 m (sd 0.117 m).

Threshold Exceedances

Using More of the Extreme Observations

Let X denote an arbitrary observation from F. Choose a high threshold u and define the excess

$$Y = X - u \mid X > u.$$

Its conditional distribution function is

$$H_u(y) = P(X - u \leq y \mid X > u) = \frac{F(u+y) - F(u)}{1 - F(u)}, \qquad y \geq 0.$$

- Block maxima retain one value per block; threshold methods retain every exceedance, subject to dependence treatment.
- We must model both **how often** exceedances occur and **how large** they are.

The Generalized Pareto Model

A Second Limit Theorem with the Same Shape Parameter

If F lies in a GEV domain of attraction with shape ξ, then for a sufficiently high threshold u,

$$H(y) = 1 - \left(1 + \frac{\xi y}{\sigma_u}\right)^{-1/\xi}, \quad y \geq 0, \quad 1 + \xi y/\sigma_u > 0.$$

This is the generalized Pareto (GP) model for the excess $Y = X - u$.

- $\xi = 0$: $H(y) = 1 - e^{-y/\sigma_u}$ (exponential).
- $\xi < 0$: excesses have upper endpoint $-\sigma_u/\xi$.
- $\xi > 0$: excesses have a power-law tail.

The GP scale σ_u depends on the threshold; ξ agrees with the GEV shape.

Extrapolating a Tail Probability

Combine the Exceedance Rate and Excess Distribution

Let $\zeta_u = P(X > u)$. For $x > u$ within the GP support,

$$P(X > x) \approx \zeta_u \left\{ 1 + \xi \frac{x - u}{\sigma_u} \right\}^{-1/\xi}.$$

With n_u exceedances among n observations, estimate ζ_u by n_u/n. The level exceeded once every m observations is

$$x_m \approx u + \frac{\sigma_u}{\xi} \{ (m\zeta_u)^{\xi} - 1 \},$$

or $x_m \approx u + \sigma_u \log(m\zeta_u)$ when $\xi = 0$.

Key Point

The exceedance rate is part of the model. The conditional excess law alone does not determine an unconditional event probability.

A Threshold Calculation

The Units and the Conditioning Matter

Example (Illustrative Severity Model; Arbitrary Units)

Suppose $u = 10$, $\zeta_u = 0.05$, $\sigma_u = 2$ and $\xi = 0.2$. The level exceeded once every $m = 1000$ observations is

$$x_{1000} \approx 10 + \frac{2}{0.2}(50^{0.2} - 1) \approx 21.87.$$

If instead $\xi = 0$, with the same u, ζ_u, σ_u,

$$x_{1000} \approx 10 + 2 \log 50 \approx 17.82.$$

Choosing a Threshold

An Approximation–Information Trade-Off

- A low u gives more exceedances, but the GP approximation may be poor.
- A high u strengthens the asymptotic rationale, but leaves little information.

For an exact GP tail and $v > u$,

$$\xi_v = \xi_u = \xi, \qquad \sigma_v = \sigma_u + \xi(v-u) > 0.$$

Thus examine shape stability and the modified scale $\sigma_u - \xi u$. If $\xi < 1$,

$$\mathrm{E}(X - v \mid X > v) = \frac{\sigma_u + \xi(v-u)}{1-\xi}.$$

Likelihood for GP Excesses

Learning the Scale and Shape

For approximately independent excesses $y_1, \ldots, y_{n_u}$, and $\xi \neq 0$,

$$\ell(\sigma_u, \xi) = -n_u \log \sigma_u - \left(1 + \frac{1}{\xi}\right) \sum_{i=1}^{n_u} \log\left(1 + \frac{\xi y_i}{\sigma_u}\right).$$

Require $\sigma_u > 0$ and $1 + \xi y_i/\sigma_u > 0$ for every excess. For $\xi = 0$,

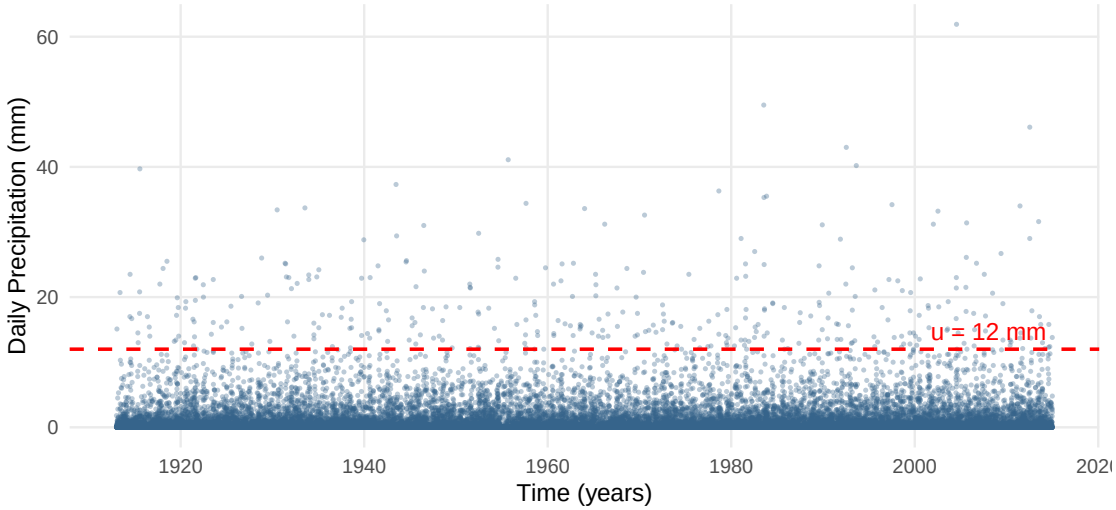
$$\ell(\sigma_u, 0) = -n_u \log \sigma_u - \sum_i y_i/\sigma_u, \qquad \widehat{\sigma}_u = \overline{y}.$$

Key Point

Parameter-dependent support must be kept in mind during numerical maximization.

Abisko Daily Precipitation

A Real GP Analysis Using `ismev2`



Data: 37,256 daily amounts, 1913–2014.

Threshold: $u = 12$ mm; 335 excesses.

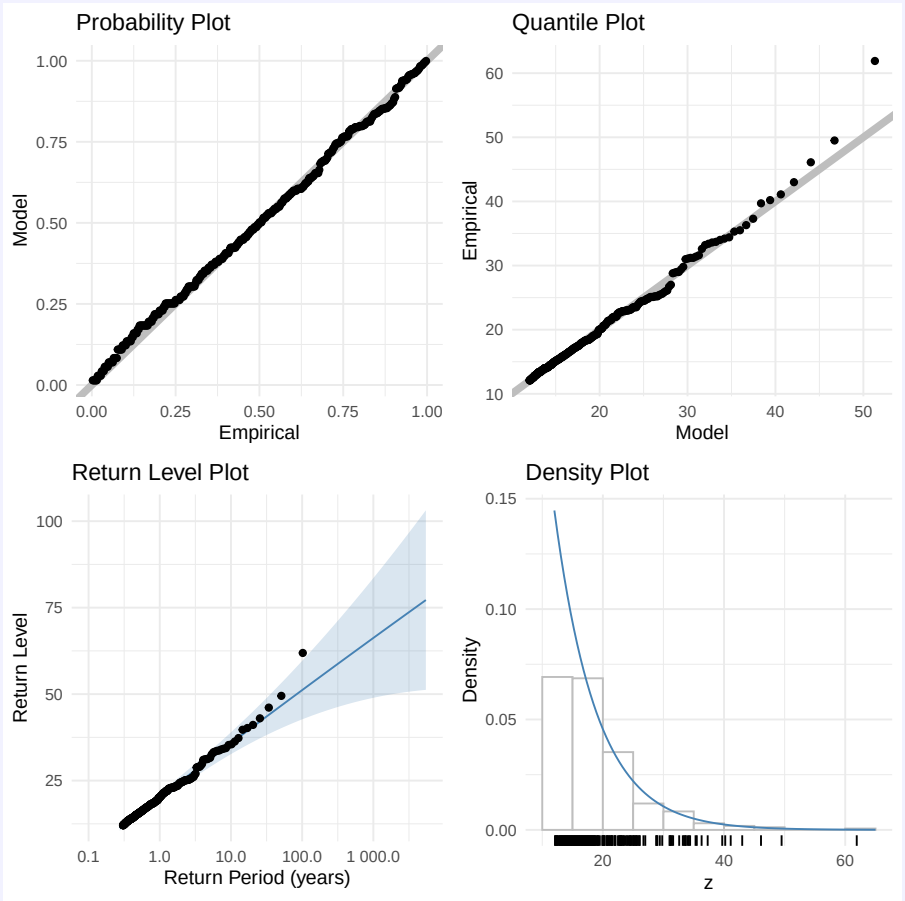
Stationary GP fit:
$\widehat{\sigma}_u = 6.912$ mm,
$\widehat{\xi} = -0.008$.

100-year daily level:
$\widehat{x}_{36500} = 51.16$ mm.

```
fit <- gpd.fit(abisko$precip, threshold = 12)
```

Abisko Daily Precipitation

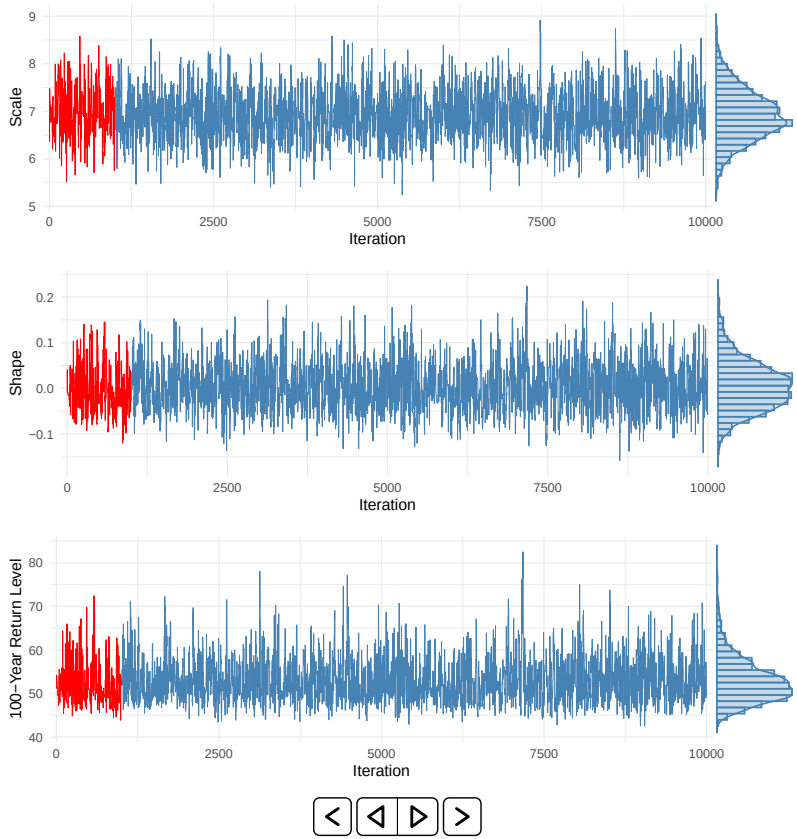
Diagnostics from `ismev2`



The largest observation exposes the uncertainty that matters most for long return periods.

Abisko Daily Precipitation

A Bayesian GP Analysis as the Markov Chain Runs



The same **GP likelihood** from `ismev2` can be combined with a prior:

$$\pi(\sigma_u, \xi \mid \mathbf{y}) \propto L(\sigma_u, \xi; \mathbf{y}) \times \pi(\sigma_u, \xi).$$

Red: first 1,000 iterations.

Blue: 9,000 retained iterations.

Transforming every retained draw gives x_{36500}:

posterior mean 52.89 mm; 95% interval $[45.50, 65.27]$ mm.

Changing Environments

Conditional Tail Models and Nonstationarity

In the local regression notation used here, a simple extension lets block-maximum parameters depend on the predictor x:

$$Y \mid X = x \sim \mathrm{GEV}\{\mu(x), \sigma(x), \xi(x)\}.$$

For example,

$$\mu(x) = b_0 + b_1 x, \qquad \log \sigma(x) = a_0 + a_1 x, \qquad \xi(x) = \xi_0.$$

- Splines or neural networks can replace linear predictors.
- Flexible shape functions need substantial tail information and regularization.
- Under change, report probabilities conditional on a specified value of x.

Extremes Can Cluster over Time

Marginal Tail Behaviour Does Not Determine Event Frequency

Storms can produce several consecutive threshold exceedances. For a stationary sequence, under suitable dependence conditions,

$$n\{1 - F(u_n)\} \to \tau \quad \Longrightarrow \quad P(M_n \leq u_n) \to e^{-\theta\tau},$$

where $\theta \in (0, 1]$ is the **extremal index**.

- $\theta = 1$ for iid observations; values below one reflect clustering in this limit.
- Use declustering or an explicit dependence model when appropriate.
- Distinguish numbers of exceedance observations from numbers of extreme episodes.

What Makes an Event Multivariate?

Define the Risk Region Before Fitting a Model

For a pair (X, Y), different events answer different questions:

$$\begin{aligned} \text{Both high:} &\quad \{X > x,\ Y > y\}, \\ \text{At least one high:} &\quad \{X > x \text{ or } Y > y\}, \\ \text{Large aggregate:} &\quad \{X + Y > s\}. \end{aligned}$$

Example (Two Sites with Marginal Exceedance Probability 0.01)

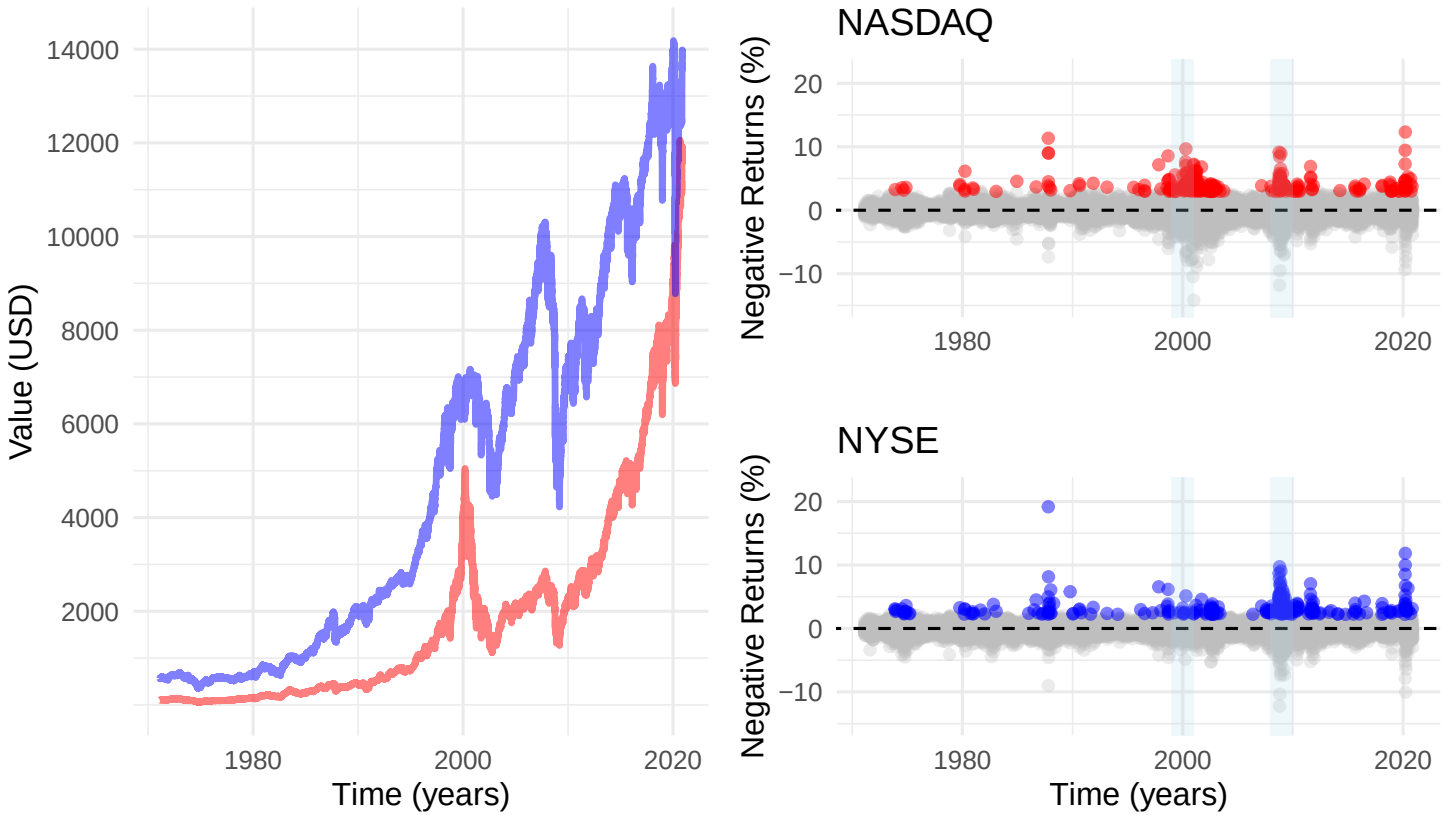
If independent, the probability that both exceed is 0.0001. If perfectly dependent with aligned quantiles, it is 0.01.

Key Point

Risk is often about joint extremes (e.g., joint losses in a portfolio).

NASDAQ and NYSE

Prices and Negative Returns



For price P_t, the daily loss is $L_t = (P_{t-1} - P_t)/P_{t-1}$. Joint extreme losses occur together; as will be seen later, their dependence can change over time.

Standardizing the Margins

Separate Marginal Tail Behaviour from Dependence

For an original pair (Z_1, Z_2) with continuous marginal distribution functions F_1, F_2, set

$$X = \frac{1}{-\log F_1(Z_1)}, \qquad Y = \frac{1}{-\log F_2(Z_2)}.$$

Then, in the notation used from this point onward,

$$P(X \leq x) = \exp(-1/x), \qquad P(Y \leq y) = \exp(-1/y), \qquad x, y > 0.$$

In practice, margins are often estimated by using the empirical distribution function.

Componentwise Maxima

The Two Maxima May Come from Different Storms

As the following example shows, the componentwise maximum does not need to be an observed point.

For storm i, let (X_i, Y_i) denote the standardized measurements at two sites. Across n iid storms, define

$$\mathbf{M}_n = (M_{x,n}, M_{y,n}) = (\max\{X_1, \ldots, X_n\}, \max\{Y_1, \ldots, Y_n\}).$$

Two Storms. Suppose $(X_1, Y_1) = (10, 2)$ and $(X_2, Y_2) = (3, 12)$. Then

$$\mathbf{M}_2 = (\max\{10, 3\}, \max\{2, 12\}) = (10, 12).$$

The largest X comes from storm 1, whereas the largest Y comes from storm 2. Thus $\mathbf{M}_2$ does not equal either observed pair.

A Measure-Dependent Measure

Why the Parameter Is Itself a Probability Measure

- In multivariate extremes we aim to learn the extremal dependence between X and Y.
- A special type of statistical model plays a key role.

Definition (Measure-Dependent Measure)

Let $\mathcal{F}$ be the space of all probability measures that can be defined over $(\Omega_0, \mathcal{A}_0)$. If G_H is a probability measure on $(\Omega_1, \mathcal{A}_1)$, for every $H \in \mathcal{H} \subseteq \mathcal{F}$, then we say that G_H is a measure-dependent measure. The family $\{G_H : H \in \mathcal{H}\}$ is a set of measure-dependent measures.

For bivariate extremes, $\mathscr{H}$ is the class of probability measures H on $[0,1]$ satisfying

$$\int_{[0,1]} w \, H(\mathrm{d}w) = \frac{1}{2}.$$

The Bivariate Extreme Value Distribution

Pickands' Representation

If

$$P(M_{x,n}/n \leq x, M_{y,n}/n \leq y) \longrightarrow G_H(x,y), \qquad n \to \infty,$$

then, for some $H \in \mathscr{H}$,

$$G_H(x,y) = \exp\{-V_H(x,y)\},$$

$$V_H(x,y) = 2\int_{[0,1]} \max\left\{\frac{w}{x}, \frac{1-w}{y}\right\} H(\mathrm{d}w).$$

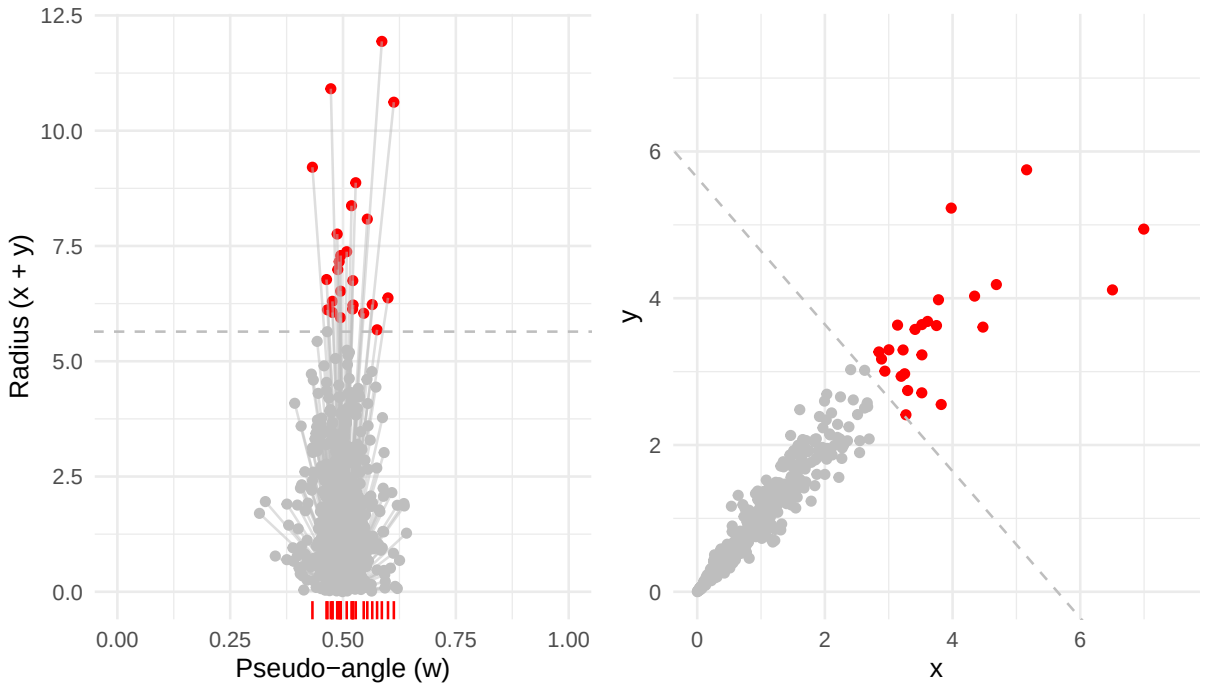
- Both the GEV and the bivariate EVD start with an "exp," but for the bivariate model $\Theta = \mathscr{H}$.
- The moment constraint is the only restriction on H; neither H nor G_H has a finite parameterization in general.
- A bivariate extreme value distribution is a measure-dependent measure.

The Geometry of Joint Extremes

Radius Measures Size; Angle Measures Relative Contribution

For a point (x, y) on the unit-Fréchet scale, Coles uses the pseudo-polar transformation

$$r = x + y, \qquad w = \frac{x}{x + y}.$$



Two Extreme Dependence Structures

The Moment Constraint Determines the Marginal Normalization

Example (Complete dependence)

$H = \delta_{1/2}$ gives

$$V_H(x, y) = \max(1/x, 1/y).$$

The two components move together.

Example (Independence)

$H = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$ gives

$$V_H(x, y) = 1/x + 1/y.$$

The joint distribution factorizes.

Both measures have mean $1/2$. Angular mass can lie on boundaries, so a smooth density is not always an adequate representation.

Estimating the Angular Measure

Statistical Problem and Inference Challenge

The angular measure H determines the interactions between joint extremes. It is the estimating target of interest.

- Statistical problem: H is unknown and is an infinite-dimensional parameter.
- Inference challenge: an estimate must obey

$$H([0,1]) = 1, \qquad \int_{[0,1]} w\, H(\mathrm{d}w) = \frac{1}{2}.$$

- It must also allow mass at 0 or 1 when the data support extremes concentrated in one component.

Threshold Pseudo-Angles

High-Radius Observations Carry Information about H

For the sample (X_i, Y_i), define

$$R_i = X_i + Y_i, \qquad W_i = \frac{X_i}{X_i + Y_i} \in [0, 1].$$

Choose a high radial threshold u; this yields $k = \sum_{i=1}^{n} I(R_i > u)$ retained pseudo-angles.

$$P(W \leq w \mid R > u) \longrightarrow H([0, w]), \qquad u \to \infty.$$

A higher u strengthens the asymptotic motivation but leaves fewer angles for inference.

The Euclidean Likelihood Angular Measure

Euclidean Likelihood Enforces the Moment Constraint

From the retained pseudo-angles $W_1, \ldots, W_k$, define

$$\widehat{H}(w) = \sum_{i=1}^{k} \widehat{p}_i I(W_i \leq w), \qquad w \in [0,1],$$

where

$$\widehat{p}_i = \frac{1}{k}\left\{1 - (\overline{W} - 1/2)S^{-2}(W_i - \overline{W})\right\}.$$

Here $\overline{W} = k^{-1}\sum_i W_i$ and $S^2 = k^{-1}\sum_i (W_i - \overline{W})^2$. The weights maximize the Euclidean loglikelihood

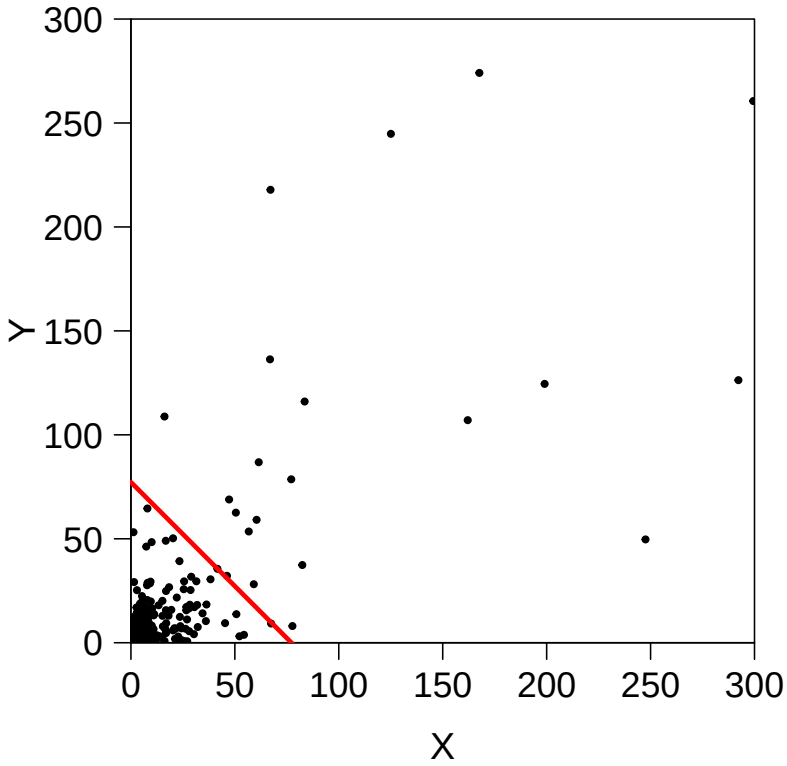
$$-\frac{1}{2}\sum_{i=1}^{k}(kp_i - 1)^2 \quad \text{subject to} \quad \sum_{i=1}^{k} p_i = 1, \qquad \sum_{i=1}^{k} W_i p_i = \frac{1}{2}.$$

Key Point

The constraints enforce total mass one and the marginal moment condition. The closed-form weights can be negative in finite samples.

Radial Threshold Exceedances

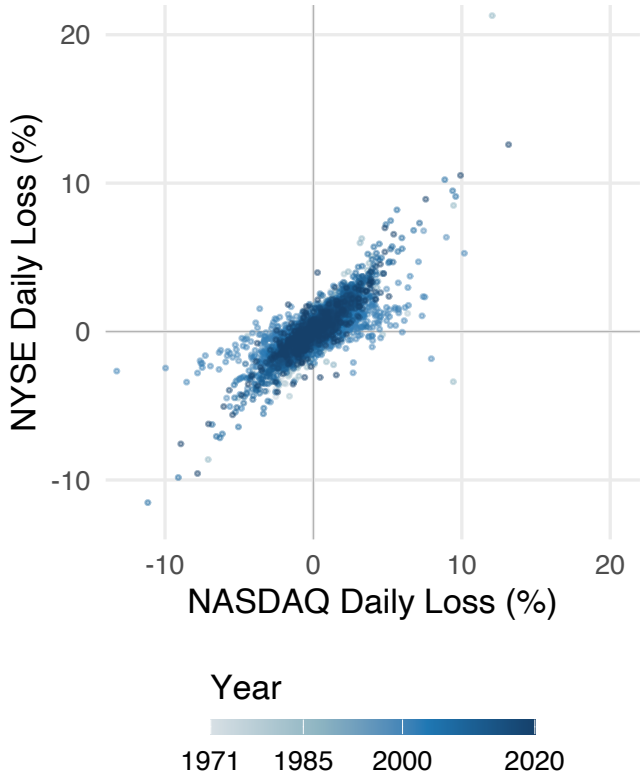
Retain $R = X + Y > u$, with $u = \widehat{q}_{0.975}(R) \approx 77.22$



The plot uses $n = 1000$ draws from a logistic model ($\alpha = 0.5$) with unit-Fréchet margins; the empirical 97.5% radial quantile retains 25 observations.

NASDAQ and NYSE Daily Losses

A Long Record Motivates Time-Varying Dependence



Data: 12,561 paired daily losses, 1971–2020.

Colour: time, from pale to dark blue.

Question: How does the extremal dependence between joint losses evolve over time?

Predictor-Dependent Angular Measures

The Dependence Structure May Vary with a Predictor

Definition (Predictor-Dependent Angular Measures)

The family $\{H_t : t \in \mathcal{T}\}$ is a set of predictor-dependent angular measures if

$$H_t \in \mathscr{H}, \qquad t \in \mathcal{T}.$$

Thus, for every fixed t,

$$H_t([0,1]) = 1, \qquad \int_{[0,1]} w\, H_t(\mathrm{d}w) = \frac{1}{2}.$$

- t may represent time, location or another predictor.
- The family describes how extremal dependence changes with t.
- Endpoint atoms remain possible for every t.

Predictor-Dependent Bivariate Extremes

From H to a Family $\{H_t : t \in \mathcal{T}\}$

The family $\{G_{H_t} : t \in \mathcal{T}\}$ is defined by

$$G_{H_t}(x, y) = \exp\left\{-2\int_{[0,1]} \max\left(\frac{w}{x}, \frac{1-w}{y}\right) H_t(\mathrm{d}w)\right\},$$

for $x, y > 0$ and $t \in \mathcal{T}$.

Univariate change

$$G_{\mu(t),\sigma(t),\xi}(z)$$

indexes location and scale by t.

Dependence change

$$G_{H_t}(x, y)$$

indexes the angular measure by t.

Key Point

G_{H_t} is both measure-dependent and predictor-dependent.

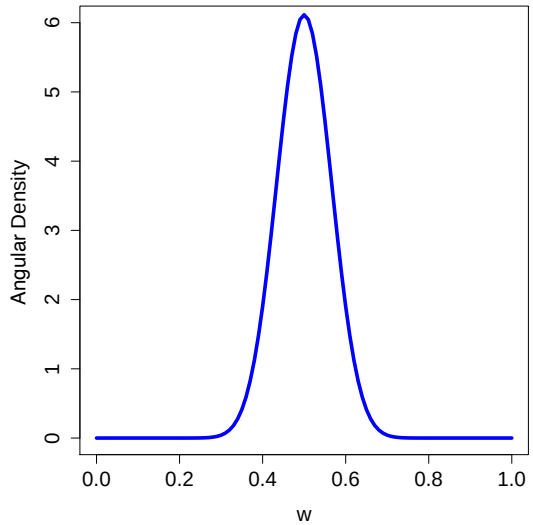
Angular Surfaces

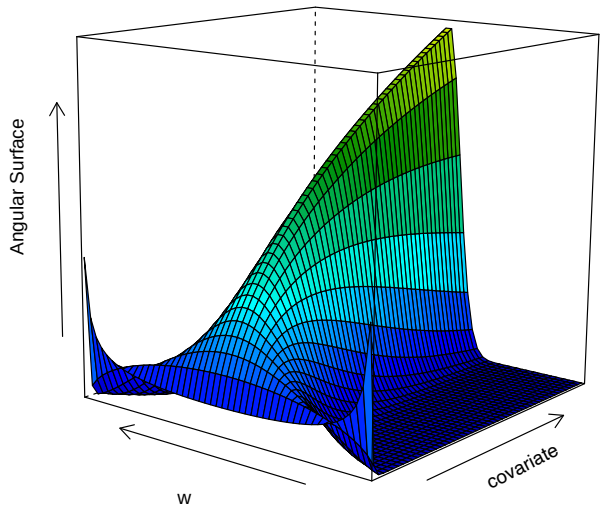
From an Angular Density to an Angular Surface

Suppose $H_t \in \mathscr{H}$ is absolutely continuous. The predictor-dependent angular density is

$$h_t(w) = \frac{\mathrm{d}H_t}{\mathrm{d}w}(w),$$

and $\{h_t(w) : w \in [0,1],\ t \in \mathcal{T}\}$ is an angular surface.





The Time-Varying Logistic Model

From a Probability-Normalized Angular Measure to Joint Risk

For $0 < \alpha_t < 1$, let H_t have density

$$h_t(w) = \frac{1}{2}\left(\frac{1}{\alpha_t} - 1\right)\{w(1-w)\}^{-1-1/\alpha_t} \times \{w^{-1/\alpha_t} + (1-w)^{-1/\alpha_t}\}^{\alpha_t - 2}, \qquad 0 < w < 1.$$

The induced bivariate extreme value distribution and tail-dependence coefficient are

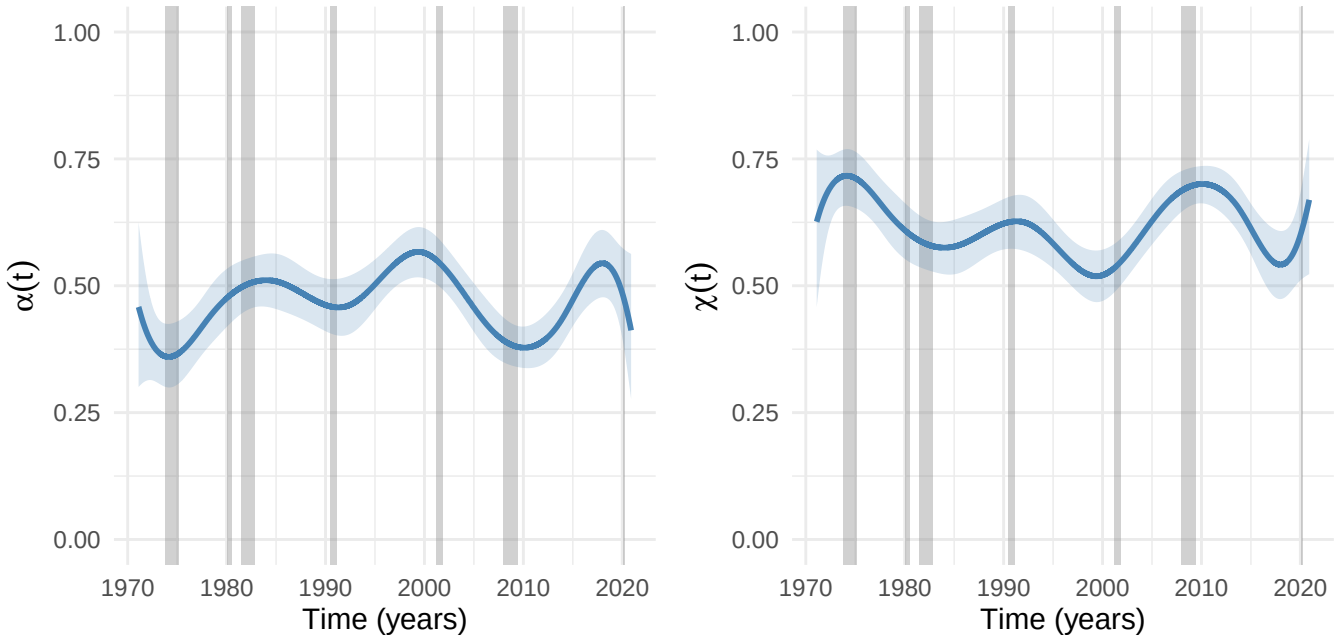
$$G_t(x,y) = \exp\left[-\{x^{-1/\alpha_t} + y^{-1/\alpha_t}\}^{\alpha_t}\right], \qquad \chi(t) = 2 - 2^{\alpha_t}.$$

Key Point

Smaller α_t means stronger extremal dependence: $\alpha_t \downarrow 0$ gives complete dependence, while $\alpha_t \uparrow 1$ gives independence. A spline for $\text{logit}(\alpha_t)$ allows smooth change over time.

A Logistic Fit to Market Losses

NASDAQ and NYSE, 1971–2020



12,561 GARCH-filtered and standardized losses; 628 radial exceedances. Blue ribbons: pointwise intervals; grey bands: NBER contractions.

Key Point

Dependence is moderate and changes gradually. It weakens around the dot-com period, when many NASDAQ losses were not matched by NYSE losses.

From Extreme Observations to Risk

A Basic Introduction to Extreme Value Theory

1. Define the Event

Block maxima M_n or excesses $X - u$, at the scale of the scientific question.

$\Longrightarrow$

2. Use a Tail Limit

GEV for maxima, GP for exceedances, both motivated by limit theory.

$\Longrightarrow$

3. Translate to Risk

Tail probabilities, return levels and joint extreme events, with uncertainty.

Nonstationarity enters through $Y \mid X = x$ and H_t; θ summarizes clustering; marginal standardization and angular measures describe how extremes occur together.

Key Takeaway

Limit theory structures extrapolation beyond observed data, into the tail of a distribution.

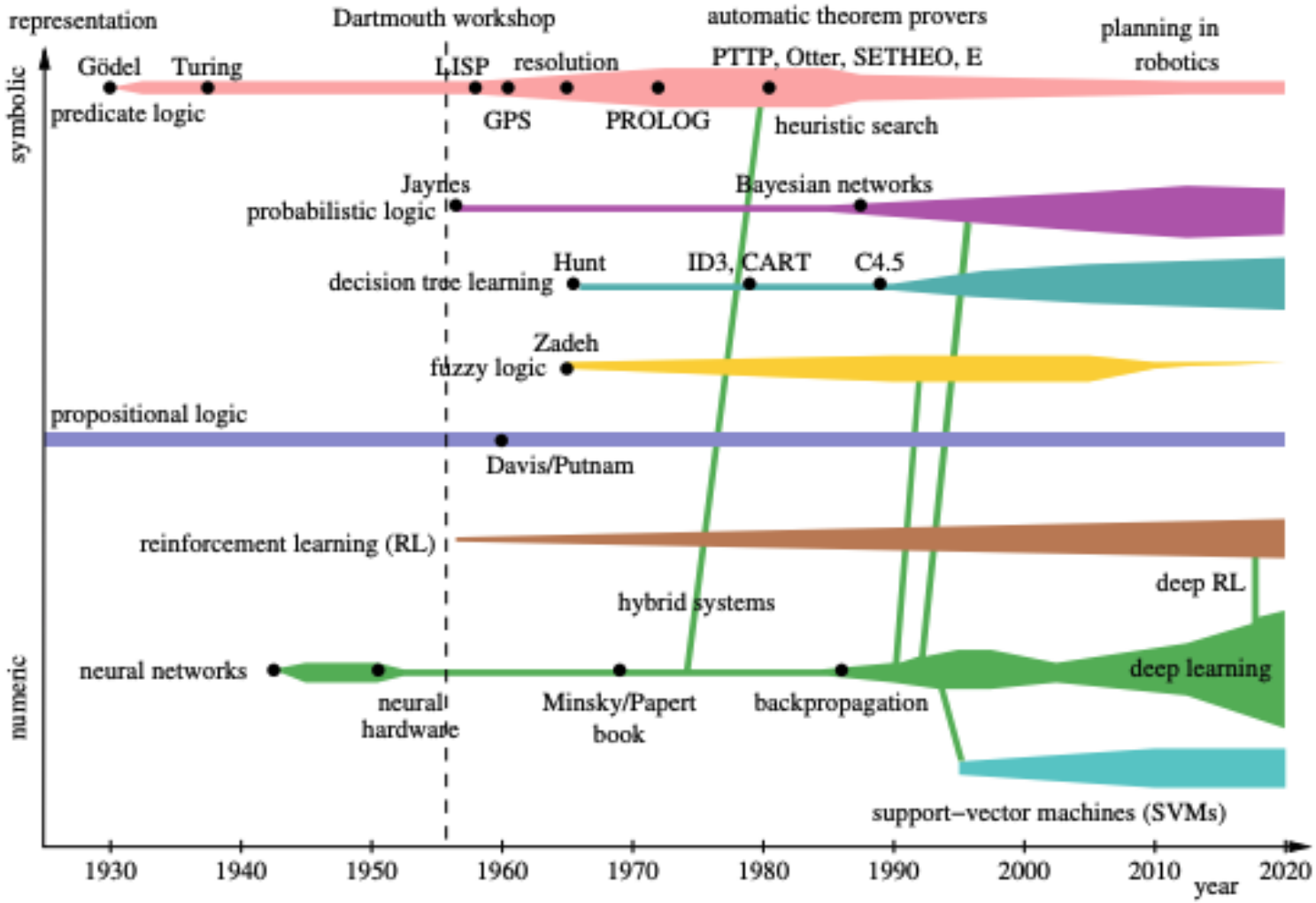
Interfaces Between Extremes and AI

An *Extremes* Special Issue

Extremes, Special Issue (2026)

Selected Comments on the History of AI

Ertel (2025)



History of various subfields of AI.

Parallel Beginnings

Fisher & Tippett (1928)

180 *Messrs Fisher and Tippett, The frequency distribution of*

Limiting forms of the frequency distribution of the largest or smallest member of a sample. By R. A. FISHER, Sc.D., Gonville and Caius College, and L. H. C. TIPPETT, M.Sc.

[*Received* 24 November, *read* 5 December, 1927.]

1. *Introductory.*

In a previous paper on the subject of the distribution of the largest member of a sample from a normal population, one of the authors has given constants involving the first four moments for samples up to 1000. In this paper, possible limiting forms of such distributions in general are discussed. It will appear that a particular group of distributions provides the limiting distributions in all cases, and that the case derived from the normal curve is peculiar for the extreme slowness with which the limiting form is approached.

2. *The possible limiting forms deduced from the functional relation which they must satisfy.*

Since the extreme member of a sample of mn may be regarded as the extreme member of a sample of n of the extreme members of samples of m, and since, if a limiting form exist, both of these distributions will tend to the limiting form as m is increased indefinitely, it follows that the limiting distribution must be such that the extreme member of a sample of n from such a distribution has itself a similar distribution.

If P is the probability of an observation being less than x, the probability that the greatest of a sample of n is less than x is P^n, consequently in the limiting distributions we have the functional equation

$$P^n(x) = P(a_n x + b_n);$$

the solutions of this functional equation will give all the possible limiting forms.

If a is not equal to unity, then

$$x = ax + b,$$

when $$x = \frac{b}{1-a},$$

and at this point $$P^n = P,$$

$$P = 0 \text{ or } 1,$$

consequently the solutions fall into three classes:

I. $a = 1$, $P^n(x) = P(x + b_n)$,

II. $P = 0$ when $x = 0$, $P^n(x) = P(a_n x)$,

III. $P = 1$ when $x = 0$, $P^n(x) = P(a_n x)$.

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Gödel (1931)

Some metamathematical results on completeness and consistency,
On formally undecidable propositions of Principia mathematica *and related systems I,*
and
On completeness and consistency

KURT GÖDEL

(*1930b*, *1931*, and *1931a*)

The main paper below (*1931*), which was to have such an impact on modern logic, was received for publication on 17 November 1930 and published early in 1931. An abstract (*1930b*) had been presented on 23 October 1930 to the Vienna Academy of Sciences by Hans Hahn.

Gödel's results are now accessible in many publications, but his original paper has not lost any of its value as a guide. It is clearly written and does not assume any previous result for its main line of argument. It is, moreover, rich in interesting details. We now give some indications of its contents and structure.

Section 1 is an informal presentation of the main argument and can be read by the nonmathematician; it shows how the argument, by dealing with the proposition that states of itself "I am not provable", instead of the proposition that states of itself "I am not true", skirts the Liar paradox, without falling into it. Gödel also brings to light the relation that his argument bears to Cantor's diagonal procedure and Richard's paradox (Herbrand, on pages 626–628 below, and Weyl (*1949*, pp. 219–235) particularly stress this aspect of Gödel's argument; see also above, p. 439).

Section 2, the longest, is the proof of Theorem VI. The theorem states that in a formal system satisfying certain precise conditions there is an undecidable proposition, that is, a proposition such that neither the proposition itself nor its negation is provable in the system. Before coming to the core of the argument, Gödel takes a number of preparatory steps:

(1) A precise description of the system P with which he is going to work. The variables are distinguished as to their types and they range over the natural numbers (type 1), classes of natural numbers (type 2), classes of classes of natural numbers (type 3), and so forth. The logical axioms are equivalent to the logic of *Principia mathematica* without the ramified theory of types. The arithmetic axioms are Peano's, properly transcribed. The identification of the individuals with the natural numbers and

Parallel Beginnings

Gnedenko (1943)

ANNALS OF MATHEMATICS
Vol. 44, No. 3, July, 1943

SUR LA DISTRIBUTION LIMITE DU TERME MAXIMUM D'UNE SÉRIE ALÉATOIRE

PAR B. GNEDENKO

(Received February 8, 1943)

Introduction

Considérons une suite

$$x_1 , x_2 , \cdots , x_n , \cdots$$

de variables aléatoires mutuellement indépendantes et assujetties à une même loi de distribution $F(x)$. Formons une autre suite de variables aléatoires

$$\xi_1 , \xi_2 , \cdots , \xi_n , \cdots ,$$

en posant

$$\xi_n = \max (x_1 , x_2 , \cdots , x_n).$$

On voit facilement que la fonction de distribution de ξ_n est

$$F_n(x) = P\{\xi_n < x\} = F^n(x).$$

L'étude de la fonction $F_n(x)$ pour les grandes valeurs de n offre un intérêt notable. Beaucoup de travaux ont été consacrés à cette question. En particulier, M. Fréchet [1] en 1927 a trouvé les lois qui peuvent être limites pour $F_n(a_n x)$ pour un choix convenable des constantes positives a_n.

Cette classe de lois limites est formée des lois de types[1] suivants

$$\Phi_\alpha(x) = \begin{cases} 0 & \text{pour} \quad x \leqq 0 \\ e^{-x^{-\alpha}} & \text{pour} \quad x > 0 \end{cases}$$

et

$$\Psi_\alpha(x) = \begin{cases} e^{-(-x)^\alpha} & \text{pour} \quad x \leqq 0 \\ 1 & \text{pour} \quad x > 0 \end{cases}$$

où α désigne une constante positive.

En 1928 R. A. Fisher et L. H. C. Tippett [2] ont établi que les lois limites pour $F_n(a_n x + b_n)$, où $a_n > 0$ et b_n sont des constantes réelles convenablement choisies, se réduisent aux lois de types $\Phi_\alpha(x)$, $\Psi_\alpha(x)$ et la loi

$$\Lambda(x) = e^{-e^{-x}}.$$

R. Misès [3] qui a commencé une étude systématique des lois limites pour la distribution du terme maximum vers 1936, a trouvé plusieurs conditions suffisantes pour la convergence des lois $F_n(a_n x + b_n)$ vers chacun des types cités

[1] En imitant A. Khintchine et P. Lévy nous appellerons type de la loi $\Phi(x)$ l'ensemble des lois $\Phi(ax + b)$ pour tous les choix possibles des constantes réelles $a > 0$ et b.

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McCulloch & Pitts (1943)

BULLETIN OF
MATHEMATICAL BIOPHYSICS
VOLUME 5, 1943

A LOGICAL CALCULUS OF THE IDEAS IMMANENT IN NERVOUS ACTIVITY

WARREN S. MCCULLOCH AND WALTER PITTS

FROM THE UNIVERSITY OF ILLINOIS, COLLEGE OF MEDICINE,
DEPARTMENT OF PSYCHIATRY AT THE ILLINOIS NEUROPSYCHIATRIC INSTITUTE,
AND THE UNIVERSITY OF CHICAGO

Because of the "all-or-none" character of nervous activity, neural events and the relations among them can be treated by means of propositional logic. It is found that the behavior of every net can be described in these terms, with the addition of more complicated logical means for nets containing circles; and that for any logical expression satisfying certain conditions, one can find a net behaving in the fashion it describes. It is shown that many particular choices among possible neurophysiological assumptions are equivalent, in the sense that for every net behaving under one assumption, there exists another net which behaves under the other and gives the same results, although perhaps not in the same time. Various applications of the calculus are discussed.

I. Introduction

Theoretical neurophysiology rests on certain cardinal assumptions. The nervous system is a net of neurons, each having a soma and an axon. Their adjunctions, or synapses, are always between the axon of one neuron and the soma of another. At any instant a neuron has some threshold, which excitation must exceed to initiate an impulse. This, except for the fact and the time of its occurrence, is determined by the neuron, not by the excitation. From the point of excitation the impulse is propagated to all parts of the neuron. The velocity along the axon varies directly with its diameter, from less than one meter per second in thin axons, which are usually short, to more than 150 meters per second in thick axons, which are usually long. The time for axonal conduction is consequently of little importance in determining the time of arrival of impulses at points unequally remote from the same source. Excitation across synapses occurs predominantly from axonal terminations to somata. It is still a moot point whether this depends upon irreciprocity of individual synapses or merely upon prevalent anatomical configurations. To suppose the latter requires no hypothesis *ad hoc* and explains known exceptions, but any assumption as to cause is compatible with the calculus to come. No case is known in which excitation through a single synapse has elicited a nervous impulse in any neuron, whereas any

Nearly a Century Later. EVT and AI's intellectual foundations emerged in the same era; today their traditions are meeting.

Next Lecture. Classical generative modeling, before EVT enters.